

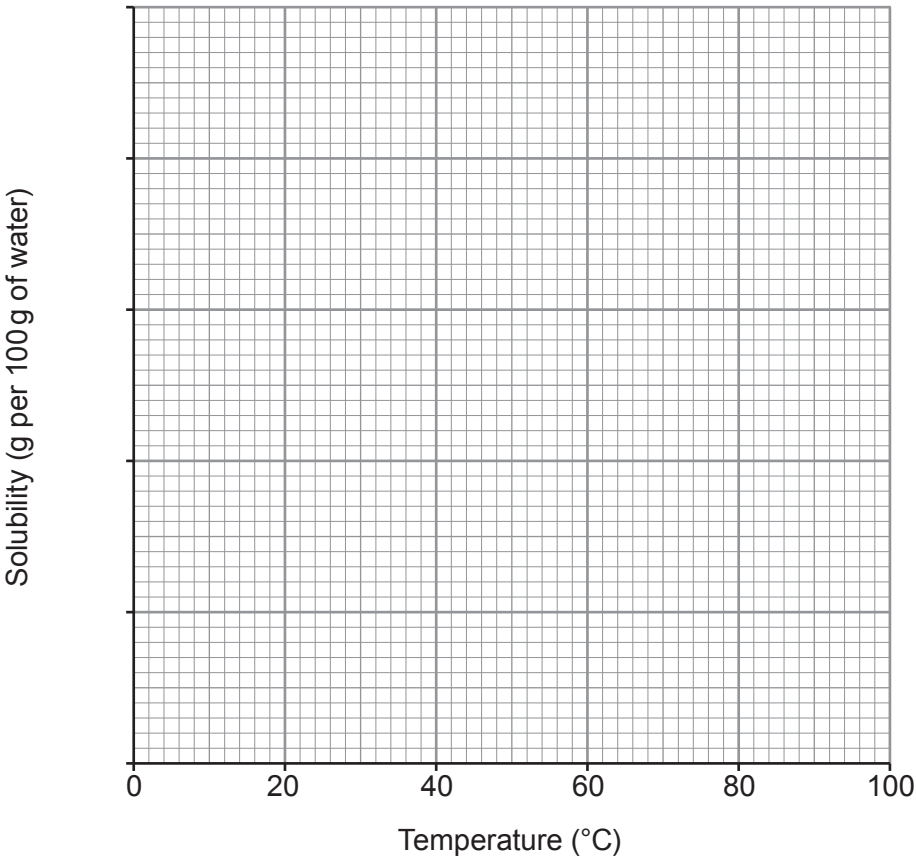
4. (a) The table shows the solubility of potassium nitrate in water at different temperatures.

| Temperature (°C) | Solubility (g per 100 g of water) |
|------------------|-----------------------------------|
| 0                | 13                                |
| 20               | 32                                |
| 40               | 64                                |
| 60               | 110                               |
| 80               | 169                               |
| 100              | 246                               |

(i) Choose a suitable scale for the y-axis and plot the data on the grid.

Draw a line of best fit.

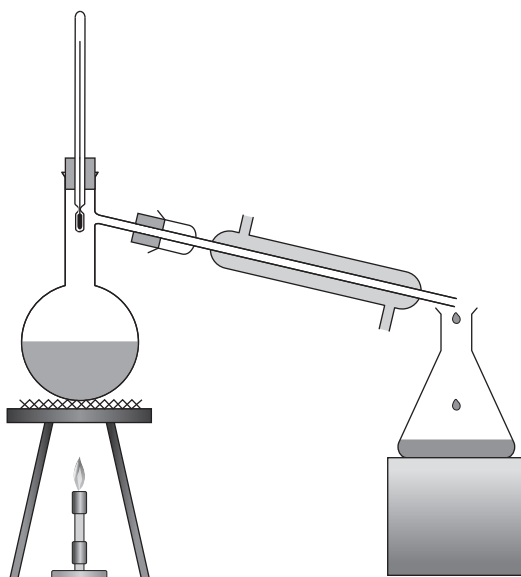
[4]



- (ii) Calculate the mass in grams of potassium nitrate that would be crystallised if  $250\text{ cm}^3$  of a saturated solution were cooled from  $80^\circ\text{C}$  to  $30^\circ\text{C}$ . [3]

Mass = ..... g

- (b) Ethanol can be separated from water by distillation using the apparatus shown.



Explain how ethanol and water are separated by distillation.

[2]

.....

.....

.....

